

Introduction to Cloud Computing

Session 1 | Student Handout

Local (On-Premise) Computing vs. Cloud Computing

Course	Cloud Computing (AWS & Microsoft Azure)
Institution	IIMT University, Meerut
Session	1 of the foundation module
Applies to	AWS batch (1st Semester) and Azure AZ-104 batch (3rd Semester)
Prerequisite	None. This session starts from zero.

What you will be able to do after this session

- Explain, in your own words, what cloud computing actually is.
- List the real costs and limits of running your own local servers.
- Compare local and cloud setups across cost, speed, scale and maintenance.
- Identify the three service models (IaaS, PaaS, SaaS) with examples.
- Identify the four deployment models (Public, Private, Hybrid, Community).
- Recognise the same concept under both AWS and Azure names.

How to use this handout

Read it once before the session and once after. The tables are the parts worth memorising. The self-check questions on the last page are what you should be able to answer without looking anything up.

1. Start With the Problem, Not the Definition

Before understanding what cloud computing is, you need to understand the problem it was invented to solve. Definitions memorised without this context are forgotten within a week.

A realistic scenario

Suppose you and two friends decide to build a website where students can check their exam results. You write the code in a week. Now you need to put it somewhere the whole world can reach. This is where the real work begins.

In the pre-cloud era, here is what you had to arrange before a single user could visit your site:

What you needed	Why	Rough cost / time
A physical server	Your code has to run on a real machine that never switches off.	Rs. 1.5 - 3 lakh, 2-3 weeks delivery
A rack and a room	Servers cannot sit on a study table. They need a secure, dust-free space.	Rs. 30,000 onwards
Cooling	Servers generate serious heat. Overheating means hardware failure.	AC running 24x7, ongoing bill
UPS and power backup	A two-minute power cut takes your website offline.	Rs. 50,000 onwards
A static IP and leased line	A normal broadband connection cannot host a public service reliably.	Rs. 5,000+ per month
A backup server	If the main server dies, everything is lost. Backup is not optional.	Double the hardware cost
A system administrator	Someone must patch, monitor and repair it at 3 AM if needed.	A monthly salary

Add it up and you are looking at several lakh rupees and roughly a month of setup work before your first visitor arrives. And two risks remain that money cannot remove:

- **If nobody visits your site**, that entire investment is wasted. The hardware still costs the same whether one person uses it or nobody does.
- **If too many people visit**, your server crashes. Buying a bigger one takes another three weeks, by which time the traffic surge is over and your users have left.

The core problem

Traditional computing forces you to buy capacity in advance, for the maximum load you might ever face, and pay for it permanently even when it sits idle. You are guessing the future and paying for the guess.

2. What Cloud Computing Is

Cloud computing means using computing resources - servers, storage, networks, databases and software - over the internet, from a provider who already owns and operates them, and paying only for what you actually use.

You do not buy the machine. You rent its capacity, by the second, and give it back when you are done. The provider (Amazon, Microsoft, Google and others) has already built the buildings, installed the hardware, arranged the power, the cooling and the network, and hired the staff. You simply use it.

The utility analogy

You do not build a power plant in your house to get electricity. You connect to the grid, use what you need, and pay according to the meter. When you leave the house for a month, the bill drops. You are not responsible for the turbines, the transmission lines or the engineers.

Cloud computing applies that same model to computing. This is why it is often described as **computing delivered as a utility**.

The same scenario, on the cloud

Return to the results website. On the cloud, you open a browser, choose a server size, and click create. In roughly ninety seconds you have a running server with an IP address. Your site goes live the same afternoon. On results day, when fifty thousand students arrive at once, you add more servers automatically and remove them the next morning. You pay for those extra servers for exactly the hours they existed.

3. A Short History

Cloud computing did not appear suddenly. It is the result of several older ideas maturing at the same time.

Period	What happened	Why it mattered
1960s	Mainframe time-sharing. Many users shared one large, expensive computer.	The first idea that computing could be shared rather than owned.
1990s	The internet became widely available and commercially usable.	Remote resources could now be reached from anywhere.
Early 2000s	Virtualization matured. One physical server could be split into many independent virtual machines.	This is the single most important technical foundation of cloud.
2006	Amazon launched S3 (storage) and EC2 (servers) as public, pay-as-you-go services.	Commercial cloud computing as we know it begins.
2008-2010	Microsoft Azure and Google Cloud entered the market.	Competition, wider service ranges, falling prices.
2010s onward	Containers, serverless computing and managed services became mainstream.	Focus shifted from managing servers to building applications.

Why virtualization is the key

Virtualization lets one physical machine behave like many separate machines. Without it, a provider would need one physical server per customer, which would be far too slow and far too expensive. Virtualization is what makes renting a server by the hour possible at all. We will study it in detail in a later session.

4. How Cloud Computing Works

The mechanism is less mysterious than the name suggests. There is no cloud in the sky - there are buildings full of computers, and you are using one of them over the internet.

Layer	What it actually is
Data centre	A physical building containing thousands of servers, with industrial power supply, backup generators, cooling systems, fire suppression and armed security.
Region	A geographic area containing multiple data centres. Providers operate regions worldwide, including several in India, so your data can stay close to your users.
Availability Zone	One or more separate data centres within a region, with independent power and network. If one zone fails, the others keep running.
Virtualization layer	Software that divides physical servers into many isolated virtual machines, so multiple customers can safely share the same hardware.
Service layer	The ready-made services built on top - virtual machines, storage, databases, networking, monitoring - which is what you actually order.

Layer	What it actually is
Access layer	How you control it all: a web console, a command line tool, or code (an API).

You never see the hardware and you never enter the building. You interact with a web page or a command line, and behind it a machine somewhere in Mumbai, Singapore or Virginia begins doing your work.

5. Local Computing vs. Cloud Computing

This is the central comparison of this session. If you remember only one page from this handout, remember this one.

Aspect	Local / On-Premise	Cloud
Ownership	You own the hardware.	The provider owns it. You rent capacity.
Upfront cost	Very high. Hardware must be purchased before anything runs.	Effectively zero. You start with an account.
Cost model	Capital expenditure. Pay once, large amount, in advance.	Operating expenditure. Pay monthly for actual usage.
Time to start	Weeks to months - procurement, delivery, installation, configuration.	Minutes. A server can be running in under two minutes.
Scaling up	Buy and install more hardware. Slow and permanent.	Change a setting or let it scale automatically. Immediate.
Scaling down	Practically impossible. The hardware is already bought.	Delete what you do not need. Billing stops.
Maintenance	Entirely yours - patching, replacing failed disks, cooling, power.	The provider handles the physical layer and much of the software layer.
Backup and recovery	You must design, buy and test it yourself.	Built-in services, usually one configuration away.
Uptime responsibility	Yours. A power cut or a dead disk is your problem at any hour.	Shared. The provider commits to a service level agreement.
Access	Usually limited to your office network.	Available from anywhere with internet access.
Global reach	Requires building or renting in each country.	Deploy in another country by selecting a different region.
Data control	Complete. The data physically sits with you.	Data sits with the provider, in a region you choose.
Internet dependency	Works on the local network even if internet is down.	No internet, no access.
Best suited to	Stable, predictable workloads and strict regulatory environments.	Variable traffic, fast-moving projects, startups, global services.

An important correction

Cloud is not automatically cheaper. For a workload that runs at a steady, predictable load twenty-four hours a day for five years, owning the hardware can cost less. Cloud wins decisively when demand is uncertain, uneven or growing - which describes most modern applications, but not all of them. A good engineer knows the difference.

6. The Five Defining Characteristics

A service is genuinely cloud computing only if it has these five properties. This list comes from the widely used NIST definition and is standard across the industry.

Characteristic	Meaning	In practice
On-demand self-service	You provision resources yourself, whenever you want, without contacting a salesperson.	You click create at 2 AM and the server exists.
Broad network access	Resources are reachable over the network from standard devices.	You manage a server in Mumbai from a laptop in Meerut.
Resource pooling	The provider serves many customers from a shared pool of hardware.	Your virtual machine and another company's may sit on the same physical host, fully isolated.
Rapid elasticity	Capacity can grow and shrink quickly, often automatically, with demand.	Two servers at night, forty during the result announcement, two again by morning.
Measured service	Usage is metered and billed precisely, and you can see the meter.	You are charged for the seconds a server ran, not the month it existed.

7. Service Models: How Much Do You Manage?

Cloud services differ in how much of the stack you control and how much the provider handles. The more the provider manages, the less flexibility you have - and the less work you do.

The pizza analogy

Model	Pizza equivalent	What you manage	What the provider manages
On-Premise	You buy flour, make the dough, bake it in your own oven, eat at your table.	Everything - hardware, OS, runtime, application, data.	Nothing. There is no provider.
IaaS	You buy a frozen pizza. The oven, table and electricity are yours.	Operating system, security patches, runtime, application, data.	Physical servers, storage, network, virtualization.
PaaS	Pizza is delivered hot. You provide only the plate and table.	Your application code and your data. Nothing else.	Everything below your code, including the operating system.
SaaS	You eat at the restaurant. You bring nothing.	Only your data and your settings.	The entire stack, including the software itself.

Real examples

Model	AWS	Azure	Everyday example
IaaS	EC2 (virtual servers), EBS (disks), VPC (networks)	Virtual Machines, Managed Disks, Virtual Network	Renting an empty flat - walls and plumbing provided, furniture yours.
PaaS	Elastic Beanstalk, RDS, Lambda	App Service, Azure SQL Database, Azure Functions	A serviced apartment - furnished, maintained, you just live there.
SaaS	Amazon WorkMail, Amazon Chime	Microsoft 365, Outlook, Teams	A hotel room - everything is done for you.

Two further models appear in your syllabus and are worth knowing by name:

- **CaaS (Communication-as-a-Service)** - voice, video, messaging and telephony delivered as a service. Examples: Zoom, Twilio, Amazon Connect, Microsoft Teams Phone.
- **MaaS (Monitoring-as-a-Service)** - monitoring, alerting and log analysis delivered as a service. Examples: Amazon CloudWatch, Azure Monitor, Datadog, Zabbix Cloud.

8. Deployment Models: Where Does It Live?

Service models describe *how much* the provider manages. Deployment models describe *where* the infrastructure sits and *who* is allowed to use it. These are two different questions and students frequently confuse them.

Model	Description	Advantages	Limitations	Typical user
Public Cloud	Infrastructure owned by a provider and shared among many customers over the internet.	No upfront cost, instant scale, global reach.	Less direct control; data resides with the provider.	Startups, websites, most modern applications.
Private Cloud	Cloud infrastructure dedicated to a single organisation, on its own premises or hosted.	Full control, easier regulatory compliance, data stays in-house.	High cost, you still handle capacity planning.	Banks, defence, government, hospitals.
Hybrid Cloud	A combination of private and public, connected so workloads can move between them.	Sensitive data stays private while public handles peak load.	Complex to design, secure and operate.	Large enterprises with legacy systems.
Community Cloud	Shared infrastructure for several organisations with common requirements.	Shared cost, standards suited to that sector.	Limited flexibility, requires agreement among members.	Groups of universities, hospital networks, government departments.

Why regulated industries hesitate

Banks and financial institutions are not slow to adopt public cloud because they doubt the technology. They are bound by regulations on where data may physically reside, who may access it, how long it must be retained and how it must be audited. Regulatory requirement, not technical doubt, is usually what drives the choice of deployment model.

9. Advantages and Disadvantages

Any material that lists only advantages is selling something. Both sides matter.

Advantages	Disadvantages
No large upfront investment. You can start with almost nothing and grow as revenue arrives.	Costs can escalate silently. Forgotten resources keep billing. Unmonitored cloud bills grow surprisingly fast.
Speed. Infrastructure that once took weeks now takes minutes.	Internet dependency. If connectivity fails, you lose access to everything.
Elasticity. Capacity follows demand automatically, up and down.	Vendor lock-in. Deep use of one provider's proprietary services makes moving away expensive and slow.
Reliability. Providers offer redundancy across zones and regions that few organisations could build alone.	Reduced low-level control. You cannot inspect the physical hardware or choose its exact configuration.

Advantages	Disadvantages
Global reach. Serve users in another continent by choosing a different region.	Compliance and data residency. Certain data may legally be required to remain within national borders.
Managed services. Backups, patching and monitoring are largely handled for you.	Shared security responsibility. The provider secures the infrastructure; securing your own configuration remains entirely your job.

The most common beginner mistake

Leaving resources running after finishing an experiment. A virtual machine created for a ten-minute test continues billing for weeks if you forget it. Always delete what you create. Make this a habit from your very first practical.

10. Who Uses the Cloud, and For What

Sector	Typical use
E-commerce	Handling enormous traffic spikes during festival sales without permanent hardware.
Banking and finance	Analytics, customer-facing applications and disaster recovery, often under a hybrid model.
Healthcare	Storing medical imaging, running diagnostic analysis, telemedicine platforms.
Education	Learning platforms, online examination systems, result portals with seasonal load.
Media and entertainment	Video encoding, content delivery and streaming to global audiences.
Startups	Building and launching a product with no capital tied up in hardware.
Government	Citizen service portals, digital identity systems, public data platforms.

11. The Major Cloud Providers

Provider	Notes
Amazon Web Services (AWS)	The oldest and largest public cloud. The broadest service catalogue. Studied in the first semester of this course.
Microsoft Azure	Strongest in organisations already using Microsoft products - Windows Server, Active Directory, Microsoft 365. Studied in the third semester (AZ-104).
Google Cloud Platform (GCP)	Particularly strong in data analytics, machine learning and Kubernetes.
Others	IBM Cloud, Oracle Cloud, Alibaba Cloud, and Indian providers such as those operated by Tata and Reliance for data-residency requirements.

Learning one provider well makes the second far easier. The underlying concepts are identical; mainly the names change. The table below will be useful throughout the course.

Same concept, different names

Concept	AWS name	Azure name
Virtual server	EC2 Instance	Virtual Machine
Attached disk	EBS Volume	Managed Disk
Object storage	S3 Bucket	Blob Storage (inside a Storage Account)
Private network	VPC	Virtual Network (VNet)
Firewall rules on a resource	Security Group	Network Security Group (NSG)
Users and permissions	IAM	Microsoft Entra ID with RBAC
Load balancer	Elastic Load Balancer	Load Balancer / Application Gateway
Automatic scaling	Auto Scaling Group	Virtual Machine Scale Set

Concept	AWS name	Azure name
Managed relational database	RDS	Azure SQL Database
Run code without servers	Lambda	Azure Functions
Monitoring	CloudWatch	Azure Monitor
Infrastructure as code	CloudFormation	ARM Templates / Bicep
Content delivery	CloudFront	Azure CDN / Front Door

12. Terminology You Will Hear Constantly

Term	Meaning
Region	A geographic area where a provider operates data centres, for example Mumbai or Singapore.
Availability Zone	An isolated location within a region, with independent power and networking.
Virtual Machine / Instance	A software-defined computer running on shared physical hardware.
Provisioning	The act of creating a resource.
Elasticity	The ability to add and remove capacity quickly as demand changes.
Scalability	The ability to handle growth, by adding resources or by enlarging existing ones.
High Availability	Designing a system so that it keeps working when a component fails.
Fault Tolerance	Designing a system so that a failure causes no interruption at all.
Latency	The delay between a request and its response, usually measured in milliseconds.
Pay-as-you-go	Billing based on actual consumption rather than a fixed purchase.
SLA (Service Level Agreement)	The provider's formal commitment on uptime, and the compensation if it is missed.
Shared Responsibility Model	The division of security duties: the provider secures the cloud, you secure what you put in it.
Vendor lock-in	Dependence on one provider's proprietary services, making migration costly.
API	A defined interface that lets programs control cloud resources without a human clicking buttons.

13. Self-Check Questions

Answer these from memory. If you cannot answer one, revisit the relevant section before the next session.

1. Explain cloud computing to someone who has never heard the term, without using the word cloud.
2. List four costs of running a local server that people usually forget to count.
3. Give one situation where a local setup is genuinely the better choice than cloud.
4. State the difference between elasticity and scalability.
5. Name the five defining characteristics of cloud computing.
6. Classify each of the following: Gmail, Amazon EC2, Azure App Service.
7. What is the difference between a service model and a deployment model?
8. Why might a bank choose a private or hybrid cloud over a public one?
9. Explain vendor lock-in and give one example of how it happens.
10. Under the shared responsibility model, who secures what?

14. Before the Next Session

- Create a free account with the cloud provider for your batch. Note that a debit or credit card is usually required for identity verification, though free-tier usage is not charged. If you do not have a card, speak to the instructor - student programmes exist that do not require one.
- Open the console and simply explore it. Do not create anything yet. Get familiar with where things are.
- Find the billing or cost management page and locate it before you create your first resource. Knowing where your bill lives is the habit that separates careful engineers from expensive ones.
- Write down three questions from this handout that you want answered in the next class.

The one line to carry forward

Cloud computing did not make servers disappear. It made them someone else's responsibility to own, and your responsibility only to use well.